SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1021	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	Mannepalli Satwik	**Reg. No.:**	192424339
Date:	14/10/2026	**Duration:**	60 minutes

Code of Conduct

I M.Satwik(192424339) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: M.Satwik

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. **Understand the Problem Statement**
   - Explain the problem in your own words.
   - Identify the objectives of the proposed system.
   - State the scope of the problem.
2. **Identify Key Stakeholders and Challenges**
   - Identify all the stakeholders involved in the system.
   - Explain the roles and responsibilities of each stakeholder.
   - Discuss the major challenges and issues faced by the stakeholders.
3. **Analyze the Current Approach**
   - Describe the existing system or current process.
   - Identify its strengths and limitations.
   - Analyze the problems or gaps that need to be addressed.
4. **Propose a Suitable Solution Using Software Engineering Principles**
   - Design a software-based solution for the given problem.
   - Describe the major functional modules of the proposed system.
   - Identify the functional and non-functional requirements.
   - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.
5. **Justify the Chosen Methodology/Framework**

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. **Discuss Expected Outcomes and Limitations**
   - Describe the expected benefits of the proposed solution.
   - Explain how the solution addresses the identified challenges.
   - Discuss potential risks, implementation challenges, and limitations.
   - Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.

Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

Problem Statement:

Design and develop a distributed Satellite Ground Station Telemetry& Scheduling Network that optimizes contact windows across international satellite tracking infrastructure. The system must ingest high-velocity datapoints, including orbital mechanic trajectories, satellite health logs, ground antenna alignment precision, and local atmospheric distortion indexes. It must schedule satellite-to-station pairings dynamically and adjust tracking hard wareon the fly to maximize data download throughput.

DISTRIBUTED SATELLITE GROUND STATION TELEMETRY & SCHEDULING NETWORK

Abstract and Introduction

1. Abstract

The rapid growth of Low Earth Orbit (LEO) satellite constellations has increased the demand for efficient satellite tracking and data downloading. Traditional ground station networks face scheduling conflicts, limited contact windows, atmospheric interference, and equipment failures.

The proposed Distributed Satellite Ground Station Telemetry and Scheduling Network uses real-time telemetry, intelligent scheduling, predictive maintenance, and secure data storage to improve satellite communication. It dynamically assigns satellites to available ground stations, monitors antenna health, adapts data transmission to signal conditions, and coordinates handovers between stations.

The system aims to increase data throughput, reduce communication interruptions, improve equipment reliability, and lower operating costs.

2. Introduction

LEO satellites move rapidly around Earth and remain visible to individual ground stations for limited periods. During these contact windows, ground stations must establish communication, upload commands, and download satellite data.

Manual scheduling and independent station management may waste valuable contact time. Therefore, a distributed intelligent network is needed to coordinate stations and maximize useful data transfer.

3. Problem Statement

Existing ground station networks experience scheduling conflicts, mechanical wear, changing atmospheric conditions, and difficulties managing large datasets. These challenges reduce communication efficiency and increase operating costs.

The proposed system addresses these issues through automated scheduling, continuous equipment monitoring, adaptive link management, and secure data distribution.

4. Objectives

- Optimize satellite-to-station scheduling.
- Predict antenna wear and equipment failures.
- Improve data throughput under changing atmospheric conditions.
- Enable efficient handovers between stations.
- Store and distribute satellite datasets securely.
- Reduce maintenance and operating costs.



Figure 1: Satellite communication with a ground station.

Existing and Proposed Systems

5. Existing System

Traditional systems often use predefined schedules and separate station monitoring tools. This approach becomes inefficient when many satellites compete for limited ground station resources.

Limitations:

- Scheduling conflicts and missed contacts.
- Manual monitoring and delayed fault detection.
- Reduced signal quality during poor weather.
- Communication interruptions during handovers.
- Difficulties storing and sharing large datasets.

6. Proposed System

The proposed network connects multiple ground stations through distributed software services.

Main features:

1. Intelligent contact scheduling.
2. Real-time telemetry processing.
3. Predictive equipment maintenance.
4. Adaptive communication control.
5. Automated handover coordination.
6. Secure data storage and cataloging.
7. Centralized monitoring dashboard.

7. Existing vs Proposed System

Existing system	Proposed system
Static scheduling	Dynamic optimization

Reactive maintenance	Predictive maintenance
Manual coordination	Automated handover
Separate databases	Distributed data management
Limited monitoring	Real-time dashboard

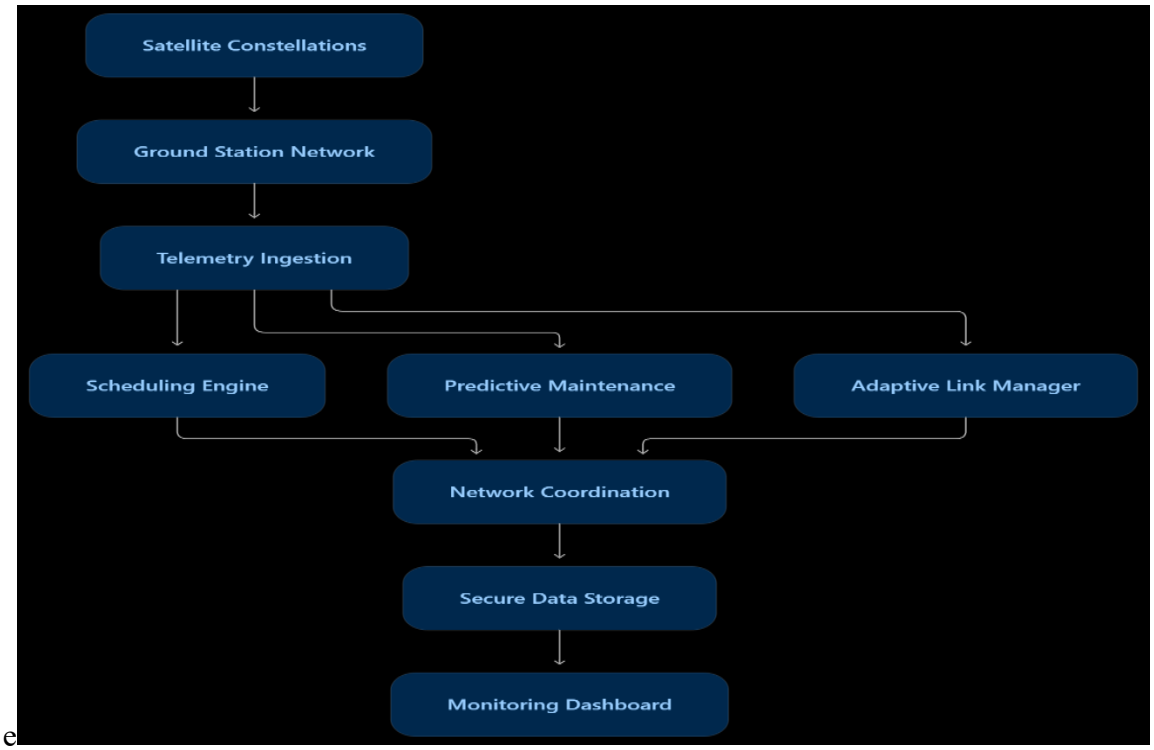
8. Scope

The system covers satellite visibility prediction, station selection, scheduling, equipment monitoring, communication optimization, and dataset management. The initial implementation can use simulated satellite trajectories and telemetry before integration with real equipment.

System Architecture

9. System Architecture

The system follows a distributed architecture in which ground stations collect telemetry and network services coordinate scheduling, analytics, and data storage.



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Figure 2: Proposed system architecture.

10. Major Modules

- Telemetry Ingestion: Collects orbital trajectories, antenna measurements, weather information, and satellite health logs.
- Scheduling Engine: Selects suitable ground stations while avoiding conflicting assignments.
- Predictive Maintenance: Detects unusual vibration, motor current, temperature, and antenna tracking errors.
- Adaptive Link Manager: Adjusts approved transmission settings according to signal quality.

- Data Management: Validates, catalogs, and securely stores downloaded data.
- Dashboard: Displays station availability, schedules, alerts, and data transfer progress.

11. Suggested Technologies

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Python, FastAPI
Scheduling	Google OR-Tools
Telemetry streaming	Apache Kafka
Database	PostgreSQL
Data storage	S3-compatible storage
Monitoring	Prometheus and Grafana
Deployment	Docker

Methodology and Scheduling Algorithm

12. Working Methodology

1. Collect satellite and ground station telemetry.
2. Predict satellite visibility windows.
3. Check antenna availability and equipment health.
4. Estimate communication quality.
5. Generate feasible station assignments.
6. Optimize the schedule.
7. Execute approved tracking commands.
8. Monitor data transfer and handovers.
9. Store verified datasets.
10. Update the dashboard and performance metrics.

13. System Workflow

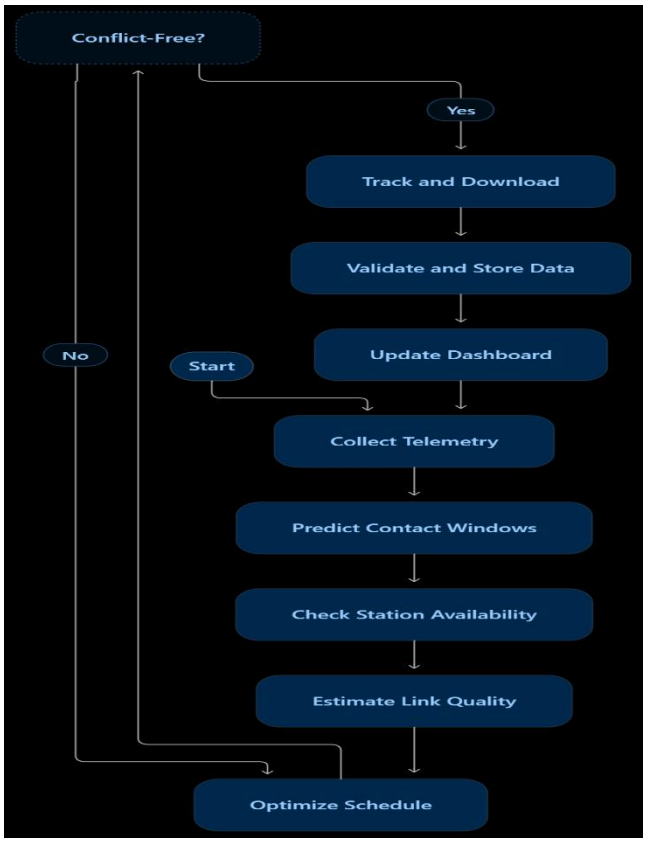


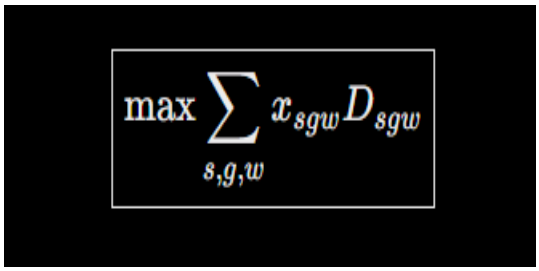
Figure 3: Overall system workflow.

14. Scheduling Algorithm

1. Start.
2. Read orbital trajectories and station availability.
3. Calculate satellite contact windows.
4. Remove incompatible or unavailable assignments.
5. Estimate expected data transfer for each contact.
6. Assign priorities to satellites.
7. Optimize station assignments.
8. Check antenna and timing conflicts.
9. Publish the schedule.
10. Monitor performance and reschedule when necessary.

15. Scheduling Objective Function

The system aims to maximize expected useful data transfer.



Where:

- \(x_{sgw}\) = 1 if a satellite is assigned to a station during a window, otherwise 0.
- \(D_{sgw}\) = expected useful data volume.
- \(s\) = satellite, \(g\) = ground station, \(w\) = contact window.

The schedule must respect visibility, antenna availability, frequency compatibility, slew time, and maintenance constraints.

Technical Feasibility

16. Meaning

Technical feasibility evaluates whether the system can be developed using available software, hardware, infrastructure, and technical expertise.

The proposed system is technically feasible as a phased project using existing software technologies and optimization algorithms. Real-world deployment requires testing and integration with station-specific equipment.

17. Framework: TELOS and Engineering Assessment

TELOS is a feasibility framework covering Technical, Economic, Legal, Operational, and Schedule dimensions. This section focuses on the technical dimension.

Criterion	Assessment
Software	Existing development tools are available
Data processing	Distributed event processing is suitable
Scheduling	Constraint optimization can resolve conflicts
Hardware integration	Possible through compatible control interfaces
Predictive maintenance	Possible using sensor data and analytics
Storage	Distributed storage supports large datasets
Scalability	Services can be scaled independently

18. Technical Feasibility Flowchart

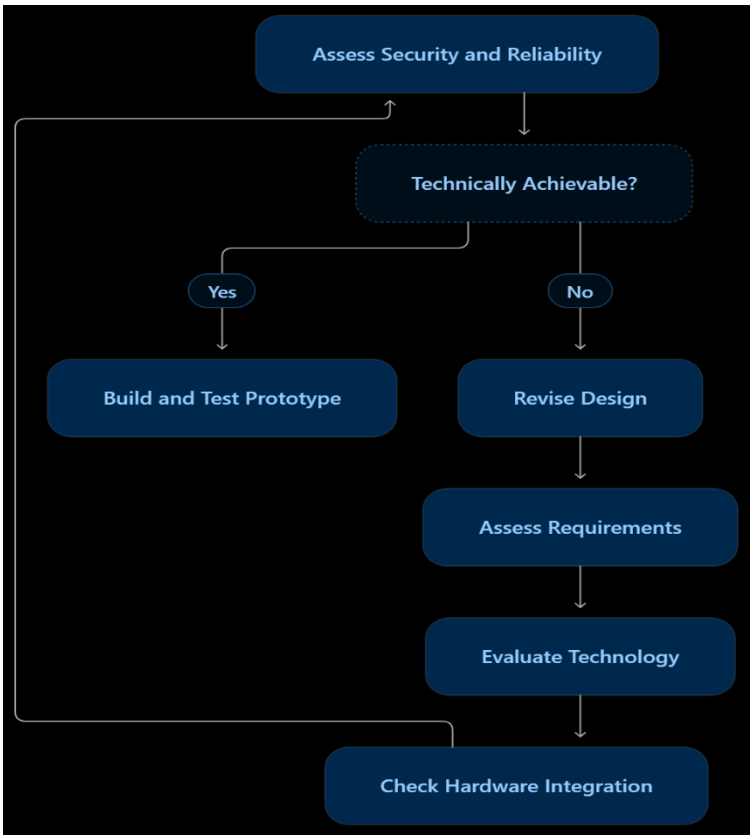


Figure 4: Technical feasibility framework.

19. Technical Risks and Solutions

- Incorrect orbital predictions: validate orbital data and update predictions.
- Excessive processing load: use scalable services and bounded optimization.
- Equipment failures: monitor sensors and implement safe fallback controls.
- Data loss: use buffering, backups, and integrity checks.
- Cybersecurity threats: apply encryption and access control.

Conclusion: The system is technically feasible if it is developed in phases, tested under simulated conditions, and integrated carefully with operational equipment.

Operational Feasibility

20. Meaning

Operational feasibility determines whether the system can function effectively in the working environment and whether operators can use it successfully.

The proposed system can simplify scheduling, improve equipment visibility, and reduce manual coordination. Successful operation requires user training and clearly defined responsibilities.

21. Framework: PIECES Analysis

PIECES evaluates Performance, Information, Economics, Control, Efficiency, and Service.

Dimension	Proposed improvement
Performance	Faster scheduling
Information	Unified telemetry and reports
Economics	Reduced avoidable maintenance
Control	Better alarms and access permissions
Efficiency	Automated station coordination
Service	Faster data delivery

22. Operational Feasibility Flowchart

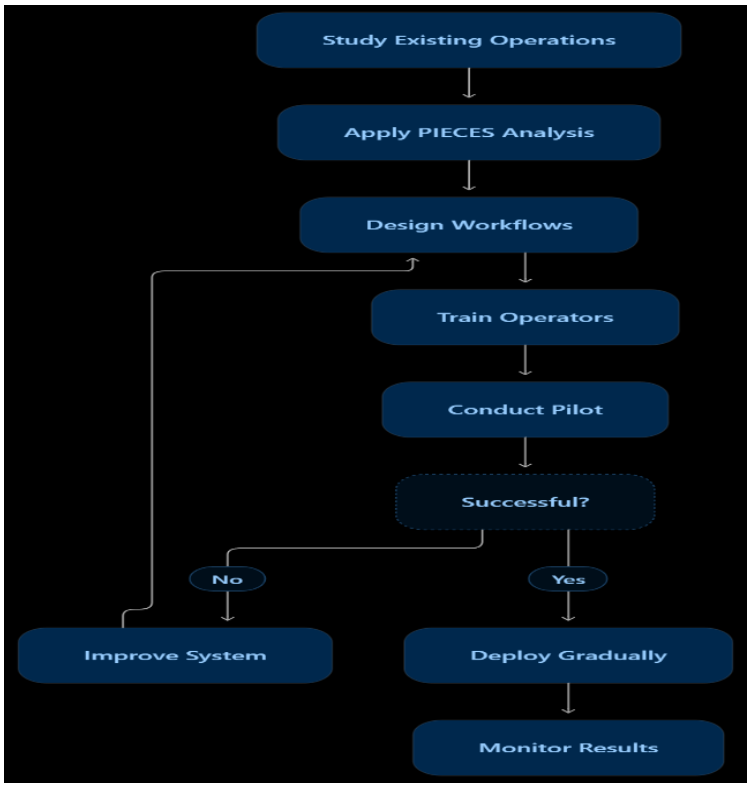


Figure 5: Operational feasibility framework.

23. Operational Roles

- Ground station operator: Monitors contacts and station health.
- Scheduler: Manages satellite priorities and contact conflicts.
- Maintenance engineer: Reviews predicted faults and schedules repairs.
- Data administrator: Manages datasets, permissions, and backups.
- System administrator: Maintains application security and availability.

24. Operational Risks

Training reduces resistance to automation. Clear alarms prevent confusion, while manual overrides and local equipment interlocks help maintain safe operations.

Conclusion: The system is operationally feasible with trained staff, clear procedures, reliable fallback mechanisms, and human supervision.

Financial Feasibility

25. Meaning

Financial feasibility determines whether the expected benefits justify the development and operating costs.

The system may reduce unnecessary maintenance, improve antenna utilization, and increase useful data downloads. However, software development, infrastructure, testing, and maintenance require investment.

26. Framework: TCO, CBA and ROI

- Total Cost of Ownership (TCO): Estimates initial and recurring costs.
- Cost–Benefit Analysis (CBA): Compares costs with expected financial benefits.
- Return on Investment (ROI): Measures return relative to investment.
- Payback Period: Estimates the time needed to recover the initial investment.

27. Illustrative Initial Cost

These are hypothetical estimates for an academic software prototype, not actual market quotations.

Expense	Estimated cost
Software development	₹8,00,000
Cloud and database setup	₹2,00,000
Testing	₹2,50,000
Training and documentation	₹50,000
Total investment	₹13,00,000

Illustrative annual operating cost: ₹5,50,000.

28. Financial Feasibility Flowchart

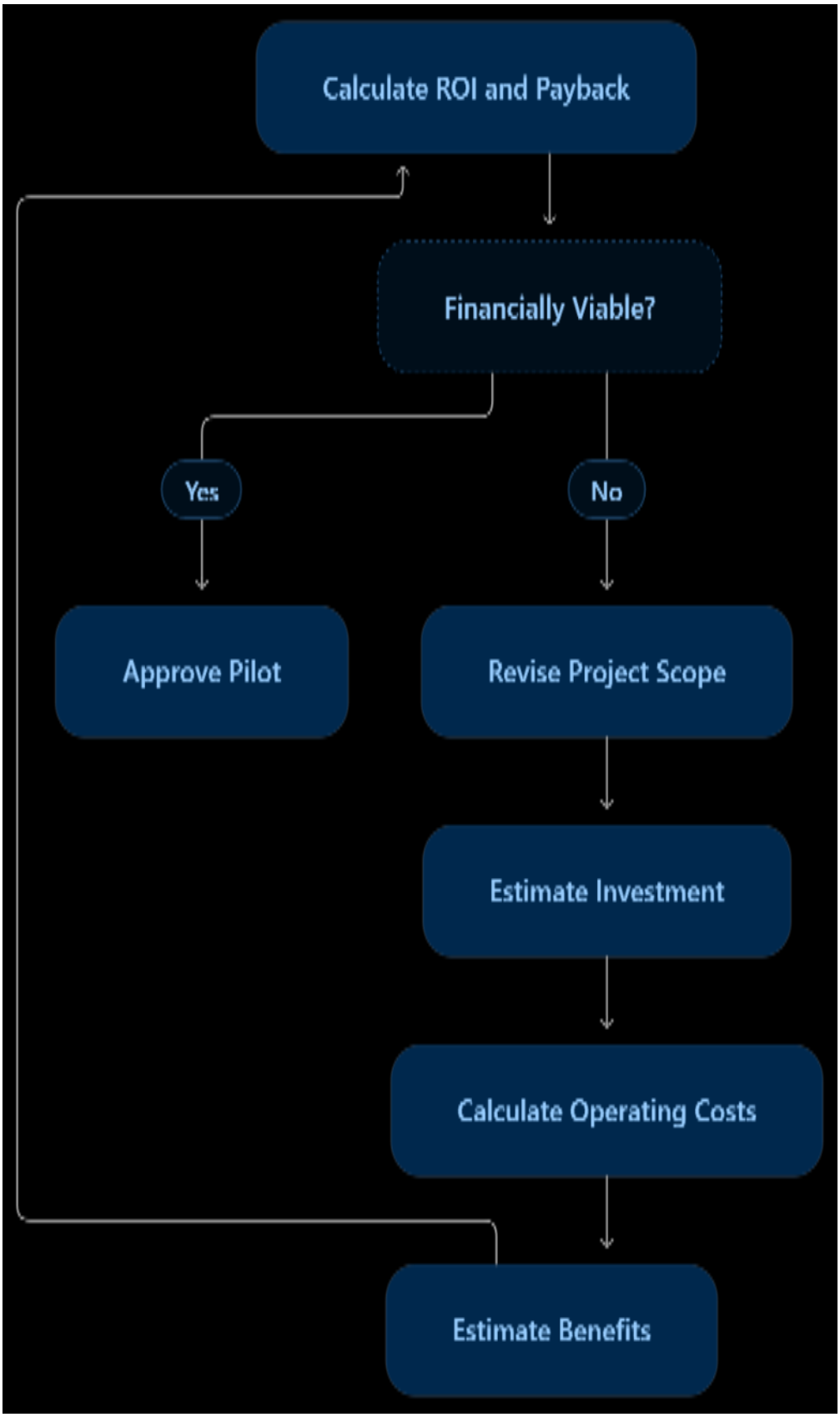


Figure 6: Financial feasibility framework.

29. ROI Calculation

Assume annual financial benefits of ₹10,00,000 and annual operating costs of ₹5,50,000.

$$\text{Net Annual Benefit} = ₹10,00,000 - ₹5,50,000$$

$$\boxed{₹4,50,000}$$

$$\text{ROI} = \frac{\text{Net Annual Benefit}}{\text{Initial Investment}} \times 100$$

$$\boxed{\text{ROI} \approx 34.62\%}$$

The simplified payback period is:

$$\frac{₹13,00,000}{₹4,50,000} \approx 2.89 \text{ years}$$

These results depend on hypothetical benefits and must be validated using actual operational data.

Conclusion: The system may be financially feasible if verified savings and additional service value exceed its lifecycle costs.

Predictive Maintenance and Adaptive Communication

30. Predictive Maintenance

Predictive maintenance identifies possible equipment problems before they cause failures.

The system monitors motor current, vibration, temperature, antenna position, and tracking error. Abnormal measurements generate alerts for maintenance engineers.

Example: Increasing vibration and motor current may indicate bearing wear. The system can recommend an inspection before a critical satellite pass.

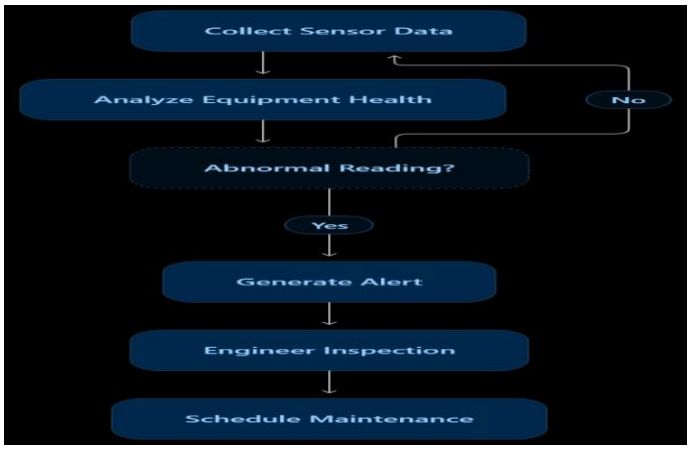


Figure 7: Predictive maintenance process.

31. Adaptive Data Transfer

Atmospheric conditions can weaken radio signals and reduce the achievable data rate. The system monitors signal strength, signal-to-noise ratio, bit-error rate, and packet loss.

When link quality decreases, the system may select a more robust approved communication profile or prioritize important data. When conditions improve, it can return to a higher-rate profile.

32. Adaptive Communication Flowchart

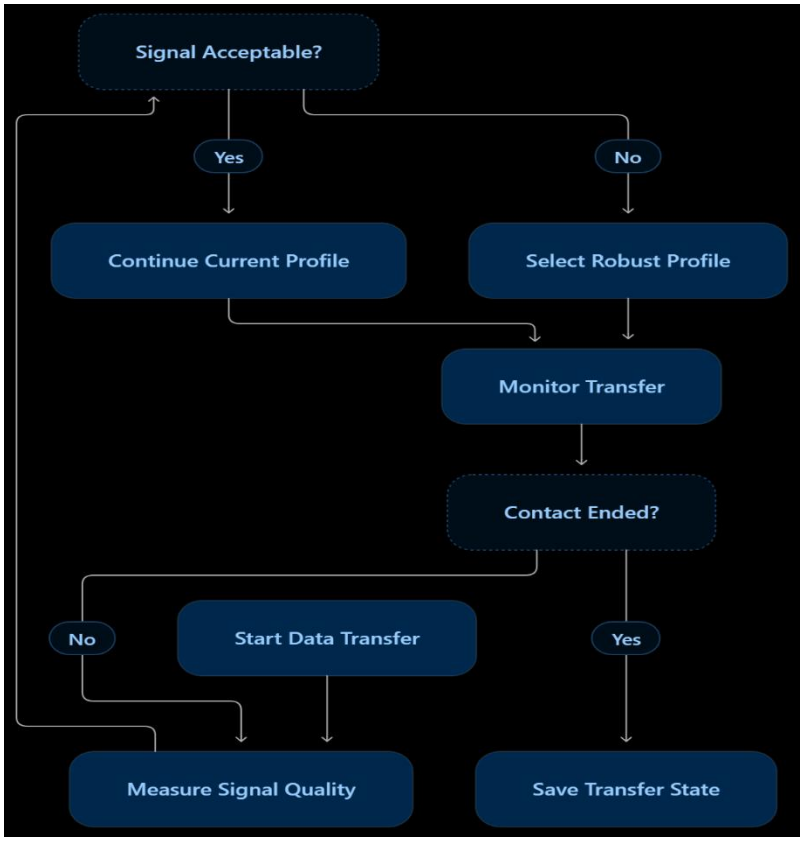


Figure 8: Adaptive communication workflow.

33. Ground Station Handover

When a satellite leaves one station's visibility region, another station may take over communication.

The system predicts the next contact, reserves a compatible station, verifies readiness, and transfers the session when authorized. Transfer recovery mechanisms help prevent loss of downloaded data.



Figure 9: Satellite communication handover between ground stations.

Security, Testing and Expected Outcomes

34. Data Security

The system protects satellite data through:

- Authentication: Verifies user and service identities.
- Access control: Restricts data and command access.
- Encryption: Protects data in transit and at rest.
- Integrity checking: Detects corrupted files using cryptographic hashes.
- Backup and recovery: Supports restoration after failures.
- Audit logging: Records important actions and system events.

35. Testing Plan

Test	Purpose
Unit testing	Verify individual functions
Integration testing	Check communication between modules
Load testing	Measure performance under high telemetry volume
Scheduling testing	Verify conflict-free assignments
Security testing	Identify access-control weaknesses
Recovery testing	Test restoration after failures
User acceptance testing	Confirm usability for operators

36. Key Performance Indicators

- Useful data throughput.
- Contact utilization.
- Scheduling conflict rate.
- Successful contact rate.
- Handover success rate.
- Equipment downtime.
- Data integrity rate.
- Operating cost per gigabyte.

37. Expected Outcomes

1. Increased useful data downloads.
2. Reduced avoidable communication interruptions.
3. Lower emergency maintenance costs.
4. Better utilization of available antennas.
5. Faster satellite data delivery.

6. Improved monitoring and operational visibility.

38. Limitations and Future Enhancements

Initial testing may use simulated data, which cannot fully represent real orbital conditions, weather, or equipment behavior.

Future development can include machine-learning-based fault prediction, advanced scheduling algorithms, digital twins, real-time weather integration, and multi-region disaster recovery.

Implementation Plan and Conclusion

39. Implementation Plan

PHASE	ACTIVITY
1	Requirement analysis
2	Architecture and database design
3	Telemetry ingestion and simulation
4	Scheduling and dashboard development
5	Maintenance and communication optimization
6	Testing and documentation

40. Feasibility Summary

Feasibility	Framework	Result
Technical	TELOS and engineering assessment	Feasible with careful integration
Operational	PIECES	Feasible with training and human supervision
Financial	TCO, CBA and ROI	Conditionally feasible, subject to verified benefits

41. Conclusion

The Distributed Satellite Ground Station Telemetry and Scheduling Network provides an intelligent approach to managing growing LEO satellite constellations. It combines automated scheduling, telemetry analysis, predictive maintenance, adaptive data transfer, coordinated handovers, and secure data storage.

The technical feasibility study shows that the major software components can be developed using existing technologies. The operational study highlights the importance of trained personnel and safe operating procedures. The financial study provides a framework for evaluating costs, benefits, ROI, and payback.

Overall, the proposed system has the potential to improve ground station utilization, data throughput, equipment reliability, and satellite data delivery. Its effectiveness should be confirmed through simulation, pilot testing, and measured operational results.